

DEVELOPMENT OF A THERMOELECTRIC HEAT-PUMPING SOLAR AIR HEATER

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Abstract—With the transition to environmentally green energy sources, thermoelectric (TE) devices offer a simple energy conversion and can be combined with solar air heaters (SAHs) to enhance thermal energy production. This study experimentally investigates an innovative approach for increasing a SAH's efficiency. Under outdoor conditions, a model SAH was tested with eleven (11) TEC1-12706 TE modules fastened to the backside of the absorber plate. Using photovoltaic modules, direct current was supplied to the TE modules which created a temperature difference allowing heat to flow from its hot side to its cold side. Thus, increasing heat transfer to the air passing beneath the absorber plate. Under the same outdoor conditions, the TE-SAHS performance was compared to a traditional SAH. The average efficiency of the TE-SAHS was 25%, a significant improvement over the 19% observed in a traditional SAH, resulting in a 31.5% improvement in efficiency. This research underscores the potential of thermoelectric technology in advancing solar thermal applications.

Keywords - Solar air heater, Thermoelectric coolers, Solar energy, Thermal efficiency, Heat transfer

I. INTRODUCTION

The efficiency of a SAH and its overall performance is determined by how effectively heat is removed from the absorber plate [1,2]. As a result, many designs have been implemented to help improve SAH's efficiency such as the use of multiple passes, fins, and phase-transition materials [4,5,6,7]. Duffie et al. [8] provide a comprehensive overview of traditional SAH enhancements, emphasizing the role of design modifications in reducing heat

NOMENCLATURE

DC	Direct Current
PV	Photovoltaic
SAH	Solar Air Heater
TE	Thermoelectric
TEC	Thermoelectric Cooler
TEG	Thermoelectric Generator

losses. Chabane et al. [9] explored the impact of airflow dynamics on thermal efficiency, demonstrating that increased air mass flow rates can enhance heat transfer. Despite these improvements, traditional SAHs still face challenges in maximizing efficiency. The integration of thermoelectric devices into SAHs promises a more effective solution to these challenges. Thermoelectric devices generally occur in two forms namely thermoelectric coolers (TECs) and thermoelectric generators (TEGs) [10].

TEGs operate under the seedback effect where a difference in temperature causes an electric voltage between two junctions [11,12,13]. On the otherhand, TECs use electricity to dissipate heat from a medium through a phenomenon known as the Peltier effect. The Peltier effect occurs when electric current (DC) flowing through a junction causes heat to flow from the hot side of the junction to the cold side [11]. The DC can be supplied directly from the grid or by a PV module. Depending on the current direction, TECs can either be used for heating or cooling. Thus,

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TECs have been applied in variety of applications such as cooling of electronic devices, refrigeration applications, and electricity generation [14,15,16].

Chang et al. [17] examined the thermoelectric air-cooling module's heat load and current intensity for electronic devices. Their experimental analysis indicates that TECs operate better at optimal input current (6A - 7A) and heat load (20W - 100W), offering a 40% greater COP than standard techniques such as electric fans.

Jugsujinda et al. [18] in an experiment to improve the thermal performance of a refrigerator using TECs were able to drop the cold side temperature from 30°C to 4.2°C in less than an hour and reported a COP of about 65%.

Emre askin elibol [19] in an experiment to improve the performance of a refrigerator integrated TECs with ZnO-TiO₂ nanofluid at different flow rates was able to increase the heat transfer rates from 19.510 W to 45.794 W and 13.639 W to 22.406 W, respectively, for the refrigeration system's heat-absorbing and heat-emitting sides.

Consequently, this same principle can be applied to SAHs to improve the rate at which heat is collected from the absorber plate thereby increasing its efficiency. This paper studies the improvement in the heat removal rate of SAHs offered by the use of TECs attached to underside of the absorber plate. It demonstrates how adding TE modules to a SAH greatly increases both its thermal efficiency and overall thermal performance.

II. MATERIALS AND METHODOLOGY

A. System description

The solar air heaters used in this experiment are made of a wooden casing having dimension 1500mm × 750mm × 150mm. The side walls and bottom are insulated with 25 mm-thick fiber to reduce heat losses by conduction. The absorber plate made of aluminum has a thickness of 2mm and was painted black to enhance heat absorption. The glazing is made of glass of dimension 1470mm × 720mm and thickness 4mm. The distance between the glass cover and the absorber plate is 30mm. The air channel has a dimension of 670 × 85 mm. An overview of the system is shown in Fig 1.

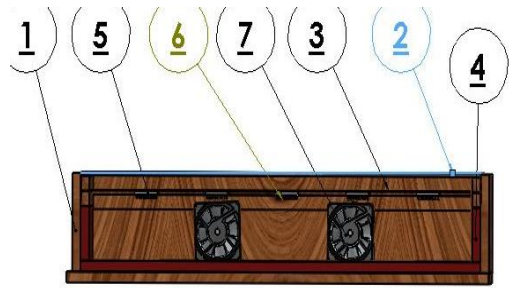


Figure 1. Front view of the SAH. 1: Casing, 2: Glazing, 3: Absorber plate, 4: Insulation, 5: TEC, 6: Heat sink (fin), 7: Fan

The absorber plate of the TE-SAH has 11 TECs attached to its backside as shown in Fig 2. Fins were added to the hot side of the TECs to increase the rate of heat flow from the TECs to the working fluid. The absorber plate of the second prototype has no TECs attached and is hence used as a conventional single-pass SAH. Two fans each were attached to the exit of the SAHs to suck hot air out of the system. The fans and TECs were powered using four (4) PV modules. Fig. 3 shows the electrical connections of the TECs and fans with the PV modules and Table 1 shows the specifications of the TECs.

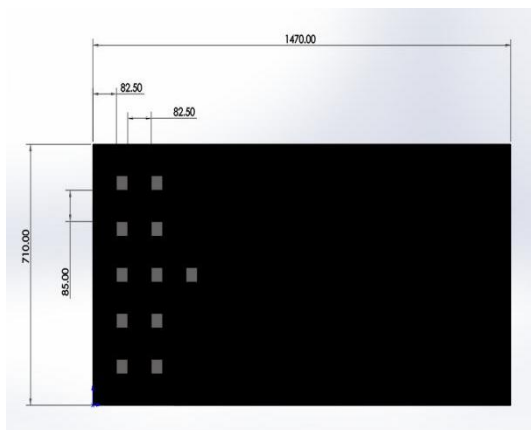


Figure 2. Absorber plate of the TE-SAH showing the position of TECs

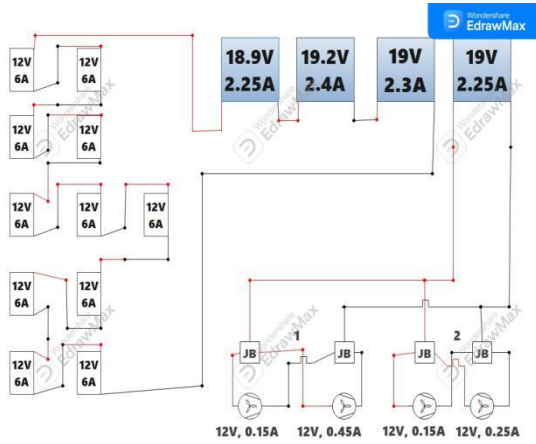


Figure 3. Circuit diagram of the fans and TECs connected to the PV modules

B. Experimental setup

The two prototypes of solar air heaters were tested in an outdoor condition at the University of Nigeria, Nsukka (6.8666° N, 7.4115° E), at an angle of 14° due north to the horizontal to enable the maximum insolation of sunlight to the system as shown in Fig 4. The experiment was run for three days, starting from 10:30 AM - 3:30 PM (5 hours). The following experimental readings were taken at 15-minute intervals each day: solar insolation, velocity of air, voltage and current across the fans and TECs, ambient temperature, and outlet temperature from the two prototype SAHs.

The thermal efficiency of the system is the ratio of the useful heat gain, $\dot{Q}_u$ to the total power input to the system.

The heat gain of the system is

$$\dot{Q}_u = \dot{m} c_p (T_{out} - T_{in}) \quad (1)$$

where $\dot{m}$ is the mass flow rate of air (kg/s), c_p is the specific heat capacity of air (1.005 KJ/kgK), T_{out} is the outlet temperature of air (°C) and T_{in} is the inlet temperature of air (°C).

The mass flow rate of air through the SAH was calculated using the following expression

$$\dot{m} = \rho A_e V_e \quad (2)$$

TABLE I. TEC SPECIFICATIONS

Characteristics	Value
Model	TEC1-12706
Dimension	40 * 40 * 3.75mm
Maximum cooling power	55W
Operating temperature	55°C - 83°C
Rated voltage	12V
Operating current	6A

where ρ is the density of air (1.127 kg/m³), A_e is the exit area in (m²) and V_e is the velocity of air at the exit (m/s).

The power input to the system was calculated using

$$\dot{Q}_s = I_s A_c + \sum IV \quad (3)$$

where I_s is the solar intensity (W/m²), A_c is the area of collector (m²), I is the current (A) and V is the voltage (V) across the DC components.

For the traditional SAH, $\sum IV$ comprises the two exit fans while for the TE-SAH, the sum includes the two exit fans as well as the TECs.

The instantaneous efficiency, η of the SAH is

$$\eta = \frac{\dot{Q}_u}{\dot{Q}_s} = \frac{\dot{m} c_p (T_{out} - T_{in})}{I_s A_c + \sum IV} \quad (4)$$



Figure 4. Experimental setup

III. RESULTS AND DISCUSSION

During testing, the TE-SAH consistently achieved higher outlet air temperatures, which translated to greater heat gain and improved overall efficiency.

Fig 5. shows the variation of solar insolation with time of day on the first day of the experiment. The average value of insolation recorded throughout the day was 278 W/m^2 . The maximum insolation recorded was 380.6 W/m^2 at 12:00 PM.

Fig 6. shows the variation of inlet (ambient) and outlet temperature of air from the two SAHs with time of day. The outlet temperature of air from the SAHs increased until 12:00 PM, when it attained its first peak. The temperature of air coming out from the traditional SAH at this point is 53°C at a mass flow rate of 0.012 kg/s and 57.5°C for the TE- SAH at the same mass flow rate.

After this point, the temperature decreases as solar insolation drops momentarily before increasing again to reach its highest point throughout the day at 1:30PM. The maximum outlet temperature recorded for the traditional SAH was 55°C at a mass flow rate of 0.01 kg/s and 61°C for the TE-SAH at a mass flow rate of 0.013 kg/s . Generally, the temperature of air coming out from the TE-SAH was $4^\circ\text{C} - 6^\circ\text{C}$ higher than the corresponding temperature for the traditional SAH, indicating a greater heat gain.

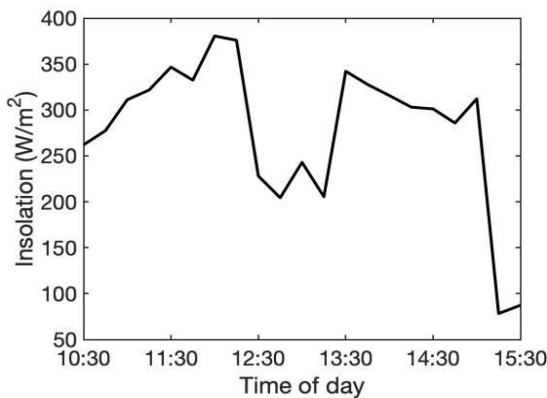


Figure 5. Variation of solar insolation with time of day on the first day of experiment

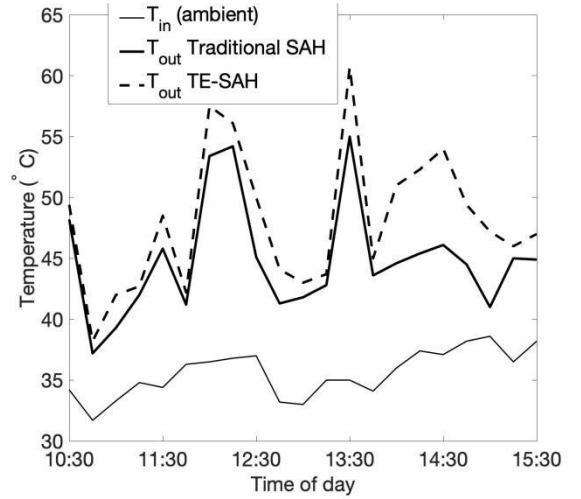


Figure 6. Variation of inlet and outlet temperatures of air with time on the first day of the experiment

The efficiency versus time graph for the two SAHs is shown in Fig 7. The efficiencies are calculated using (4). The maximum efficiency attained by the traditional SAH was 57% at 1:30PM and mass flow rate of 0.013 kg/s while the maximum efficiency for the TE- SAH was 74% at 1:30 PM and the same mass flow rate, offering an efficiency improvement of 30%. The results indicate that the TE-SAH has a better performance than the traditional SAH both in terms of heat gain (outlet temperature) and efficiency improvement.

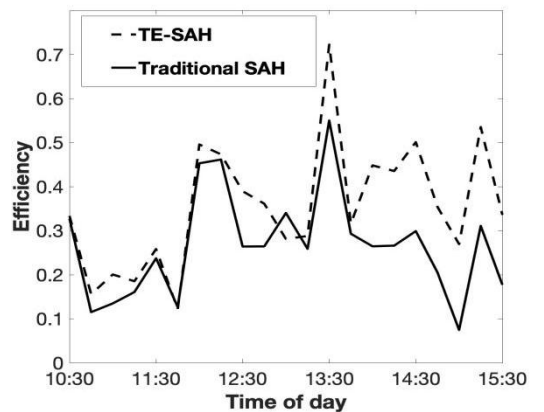


Figure 7. Variation of instantaneous efficiency of the two SAHs with time of day on the first day of experiment

Fig 8. shows the variation of solar intensity with time of day for the last day of the experiment. The solar intensity increased steadily during the morning hours reaching a peak value at midday. The maximum value of solar intensity recorded was 362 W/m^2 at 12:30PM with average insolation of 210 W/m^2 throughout the day.

Fig 9 shows the tempertaure versus time graph for the day. The maximum outlet air temperature recorded for the TE-SAH was 45.2°C compared to 43.6°C for the conventional SAH at 1:15 PM and mass flow rate of 0.01kg/s . The maximum efficiency reached by the conventional SAH was 30% at a mass flow rate of 0.011 kg/s and 36% at a mass flow rate of 0.013 kg/s for the TE- SAH, offering an efficiency improvement of about 20%. Both (maximum) efficiencies were recorded at 12:15PM as shown in Fig 10. The thermal efficiency of the TE-SAH averaged 23% over the testing period, while the conventional SAH averaged 18%, resulting in a 31.5% increase in system efficiency.

The improved performance of the TE-SAH is attributed to the improved heat transfer facilitated by the TECs. These results are supported by the collected data, which showed consistent improvements in heat gain across various operating conditions.

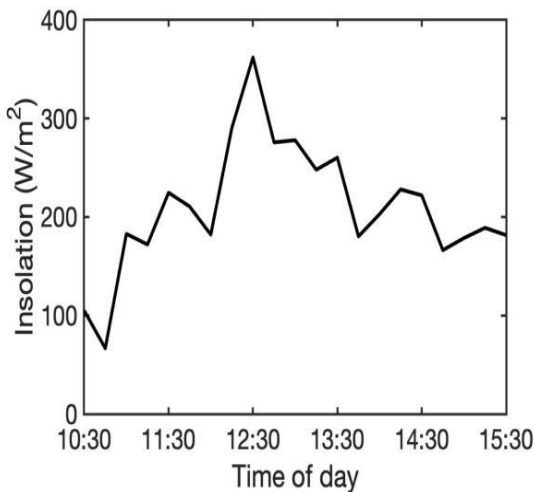


Figure 8. Variation of solar insolation with time of day on the last day of experiment

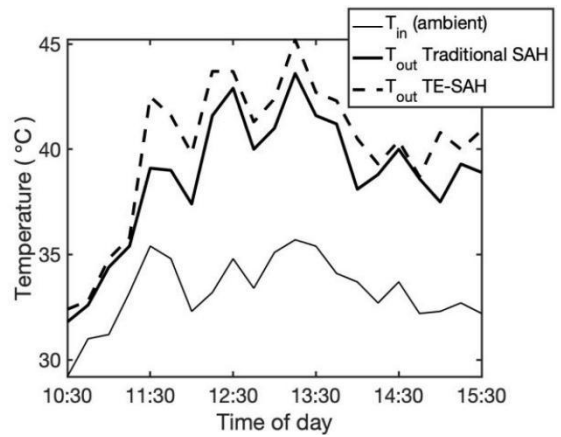


Figure 9. Variation of inlet and outlet temperatures of air from the two SAHs with time of day on the last day of experiment

By generating a temperature differential, the TECs enhanced the flow of heat from the absorber plate to the air passing through the system, leading to higher outlet air temperatures and increased heat gain. The results suggest that TEC technology offers a viable solution for improving the performance of solar thermal systems, particularly in applications where maximizing heat gain is critical. However, further research is needed to optimize the integration of TECs in SAHs, particularly concerning the balance between electrical input and thermal output.

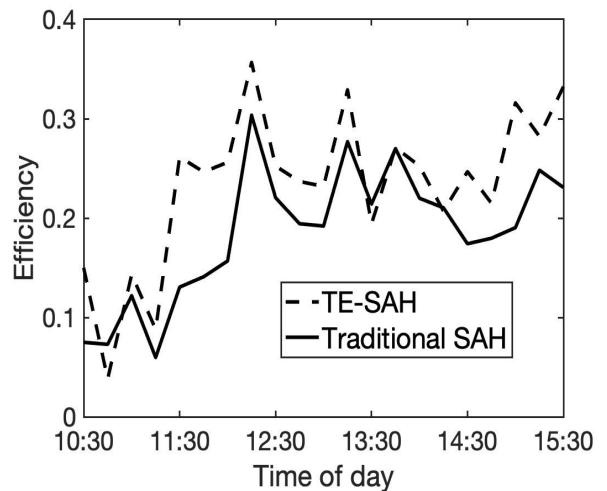


Figure 10. Variation of instantaneous efficiency of the two SAHs with time of day on the last day of experiment

IV. CONCLUSION

This study demonstrates that integrating TECs into a solar air heater can significantly enhance its thermal performance. The TE-SAH averaged 23% efficiency over the testing period, while the traditional SAH averaged 18%, resulting in a 31.5% increase in system efficiency.

A maximum efficiency of 74% was recorded for the TE-SAH at a mass flow rate of 0.013 kg/s while the maximum value for the traditional SAH is 57% at the same mass flow rate. These findings underscore the potential of TEC technology in advancing the effectiveness of solar thermal systems, offering a promising solution for various applications requiring efficient heat transfer.

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REFERENCES

- [1] Hui, X., Zhimin, D., Zhichun, L., & Wei, L. (2020). Heat transfer performance and flow characteristics of solar air heaters with inclined trapezoidal vortex generators. *Applied Thermal Engineering*, 179(115484), 1359-4311. doi:https://doi.org/10.1016/j.applthermaleng.2020.115484
- [2] Ahmed, M. Q. & Abdulmajeed, M. (2007). Analysis of high efficiency solar air heater for cold climates. *Numerical Heat Transfer Fundamentals*.
- [3] Apurba, L., Saini, J., & Solanki, S. (2007). Second law optimization of a solar air heater having chamfered rib-groove roughness on absorber plate. *Renewable Energy*, 32(10), 1967– 1980. doi: 10.1016/j.renene.2006.11.005
- [4] Abdullah, A., Abou Al-sood, M., Omara, Z., Bek, M., & Kabeel, A. (2018). Performance evaluation of a new counter flow double pass solar air heater with turbulators. *Solar Energy*, 173, 398–406. doi: https://doi.org/10.1016/j.solener.2018.07.073
- [5] Omojaro, A. & Aldabbagh, L. (2010). Experimental performance of single and double pass solar air heater with fins and steel wire mesh as absorber. *Applied Energy*, 87(12), 3759–376. doi: https://doi.org/10.1016/j.apenergy.2010.06.020
- [6] Salih, S., Jalal, M., & Najim, E. (2019). Double-pass solar air heater (DP-SAH) utilizing latent thermal energy storage. *Material Science and Engineering*. https://doi.org/10.1088/1757-899X/518/3/032038, 2019
- [7] Chabane, F., Moummi, N., Said, B., Bensahal, D., & Okba, B. (2013). Collector efficiency by single pass of solar air heaters with and without using fins. *Engineering Journal*, 17(13), 43–55. doi: 10.4186/ej.2013.17.3.43
- [8] Duffie, J.A., Beckman, W.A., & Blair, N. (2020). Solar Engineering of Thermal Processes, Photovoltaics and Wind. *John Wiley & Sons*
- [9] Chabane, F., Moummi, N., & Benramache, S. (2014). Experimental study of heat transfer and thermal performance with longitudinal fins of solar air heater. *Journal of Advanced Research*, 5(2), 183-192
- [10] Gholamreza, K., Culham, J., & Kazerouni, V. (2011). Performance analysis of multi-stage thermoelectric coolers. *International Journal of Refrigeration*, 34(8), 2129–2135. doi: 10.1016/j.ijrefrig.2011.05.015
- [11] Guclu, T. & Cuce, E. (2018). Thermoelectric coolers (TECs): From theory to practice. *Journal of electronic materials*, 48, 211 - 230. https://doi.org/10.1007/s11664-018-6753-0
- [12] Cheng-Ting, H., Gia-Yeh, H., Hsu-Shen, C., Yu, B., & Da-Jeng, Y. (2011). An effective Seebeck coefficient obtained by experimental results of a thermoelectric generator module. *Applied Energy*, 88(12), 5173-5179. https://doi.org/10.1016/j.apenergy.2011.07.033
- [13] Qasim, M., Velkin, V., & Hassan, A. (2022). Seebeck generators and their performance in generating electricity. *Journal of Operation and Automation in Power Engineering*, 10(3), 200-205. doi: 10.22098/joape.2022.9715.1677
- [14] Chein, R. & Huang, G. (2004). Thermoelectric cooler application in electronic cooling. *Applied Thermal Engineering*, 24(14), 2207–2217. doi: https://doi.org/10.1016/j.applthermaleng.2004.03.001
- [15] Yang, B., Ahuja, H., & Tran, T. N. (2008). Review Article: Thermoelectric technology assessment: Application to air conditioning and refrigeration. *HVAC&R Research*, 14(5), 635–653. https://doi.org/10.1080/10789669.2008.1039103
- [16] Alam, N., Salman Ali, M., Sajid, S., Sharma, D., & Hasan, Z. (2023). Experimental investigation and analysis of cooling performance of solar thermoelectric refrigerator. *Solar Energy*, 263(111892). https://doi.org/10.1016/j.solener.2023.111892
- [17] Chang, Y., Chang, C., Ke, M., & Chen, S. (2009). Thermoelectric air-cooling module for electronic devices. *Applied Thermal Engineering*, 29(13), 2731–2737. doi: https://doi.org/10.1016/j.applthermaleng.2009.01.004
- [18] Jugsujinda, S., Vora-ud, A., & Seetawan, T. (2011). Analyzing of thermoelectric refrigerator performance. *Procedia Engineering*, 8, 154–159. doi: https://doi.org/10.1016/j.proeng.2011.03.028
- [19] Elibol, E. A. (2024). Performance analysis of a refrigeration system integrated with a thermoelectric

cooler and microchannels in terms of heat transfer
using a hybrid nanofluid. *Applied Thermal*

Engineering, 250(123465),1359-4311.
<https://doi.org/10.1016/j.applthermaleng.2024.123465>.